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OPTIFINERS

INDUSTRIAL ENGINEERING & MANAGEMENT

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Outline of Tentative Steps

OptiProductio: Smart Production Workflow and Resource Optimization System

Phase I 29 Oct 2026	Phase II __ / __ / 2026	Phase III __ / __ / 2027
• Pick and define the problem • Literature review • SMART objectives (PO1 to PO4) • Design the 3-station line and parts list • Plan the method and timeline	• Wire sensors to Arduino UNO • Python logger and dashboard • Check counts against a hand tally • First runs and partial results • Dashboard on Bosch public data	• Rebalance and WIP-limit experiments • Results and discussion • Live prototype demo • Report, poster and PPT • IEEE-format paper draft

Status today: Phase I planning done. Hardware build in progress. Red blanks = dates not announced yet.

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EL Phase-I
Presentation Outline

- Introduction
- Literature Review
- Problem Definition
- Objectives
- Methodology and System Pipeline
- Measurement Parameters
- Research Process
- Proposed Outcomes
- Timeline of the Project
- Conclusion
- References
- Expected Outcome by the End of First Year

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INTRODUCTION

OVERVIEW OF THE TOPIC:

OptiProductio: a low-cost system that watches a small production line in real time and shows how well it is working.

How: sensors on a 3-station model line send every event (part done, part rejected, line stopped) to an Arduino UNO, which passes it to a Python dashboard on a laptop.

Why: small factories often record output by hand, so delays, pile-ups and the slowest station stay hidden until the end of the shift [2], [4].

What it adds: an alert when too many parts pile up inside the line (a WIP limit), based on the CONWIP idea [5].

Production count	**Cycle time**	**Inventory**	**Workflow**
IR sensor at each station exit	time each station takes per part	raw material bin level and WIP	bottleneck, stops and pile-ups

EL topic: Smart Production Workflow and Resource Optimization System. Deliverable: production efficiency dashboard.

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Literature Review

Key studies and findings

Study	What they did	Key finding
Hwang et al., 2017 [2]	IoT performance-measurement model built on the ISA-95 and ISO 22400 standards	IoT timestamp data can give OEE numbers in real time
Ng Corrales et al., 2020 [3]	Systematic review of OEE: 862 articles screened down to 186	Many industries adapt OEE, because the basic formula alone is not enough
Wicaksono et al., 2024 [4]	Replaced manual OEE logging with an ESP32 board over MQTT	Works in real time, but reports up to 16% OEE error
Spearman et al., 1990 [5]	Proposed CONWIP: a fixed cap on the total parts inside a line	A simple pull alternative to kanban that limits WIP
Thürer et al., 2023 [6]	Tested kanban, CONWIP and a hybrid on a physical lab-scale line	WIP limits cut the time a job spends in the line but can lower throughput

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Literature Review (contd.)

- **Trends in literature:** low-cost IoT boards now give real-time OEE [2], [4]; WIP limits like CONWIP are mostly tested in simulation or lab models [6], [9]; public line data such as the Bosch dataset is used mainly to predict failed parts [8].
- **Theoretical framework:**
  - **Little's law:** average WIP = throughput × average flow time [7]. Cut WIP at the same throughput and parts spend less time in the line.
  - **OEE = Availability × Performance × Quality** [3], [10].
  - **CONWIP:** release a new part only when a finished part leaves [5].
- **Relevance to our project:** monitoring papers report numbers but do not control the flow, and WIP-limit papers control the flow but are rarely tied to a live, measured line [6], [9]. Low-cost builds rarely check their accuracy against a manual count [4]. OptiProductio does all three on one small line.

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Problem Definition

- **Clear statement of the problem:** small production units cannot see count, cycle time, inventory and WIP while the line is running, so they find bottlenecks and pile-ups too late to act.
- **Context of the problem:** MSMEs gave 30.1% of India's GDP (2022-23) and 45.73% of its exports (2023-24) [1]. Manual OEE recording is slow because operators write data at the machine and supervisors process it later [4]. Ordinary performance systems struggle to capture what is actually happening on the floor in real time [2].
- **Significance of addressing it:** if work is released without a limit, WIP between stations can keep growing [11], parts wait longer, and the slowest station stays hidden.
- **Gap and novelty:** low-cost monitoring and WIP control are studied separately [4], [9]. OptiProductio puts both on one prototype costing under ₹2,000 (our estimate), and checks its own counts against a hand tally and Little's law [7].

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Objectives

- **Main objective:** build a low-cost hardware and software prototype that measures production count, cycle time, inventory and WIP on a 3-station model line in real time, and use it to find the bottleneck and test a WIP limit.

PO	Specific objective	Measurable target	By
PO1	Count parts at all 3 stations with IR sensors	Count error ≤ 2% against a hand tally, over three 10-minute runs	Phase II
PO2	Live dashboard: count, cycle time per station, WIP, bin level, downtime, OEE proxy	Refreshes every 2 s or faster; all 6 numbers on one screen	Phase II
PO3	Find the bottleneck and rebalance the line	Report throughput and flow time before and after, 3 runs each; Little's law gap ≤ 10%	Phase III
PO4	WIP-limit alert (LED, buzzer, dashboard)	Alert fires at the cap every time; trade-off reported for caps 2, 4, 6	Phase III

Why achievable: parts are standard IDEA Lab kit, the code is already written and tested on sample data, and each PO is checked by a simple count or timer. **Expected contribution:** a cheap, open template for monitoring and WIP control that small units and student labs can copy.

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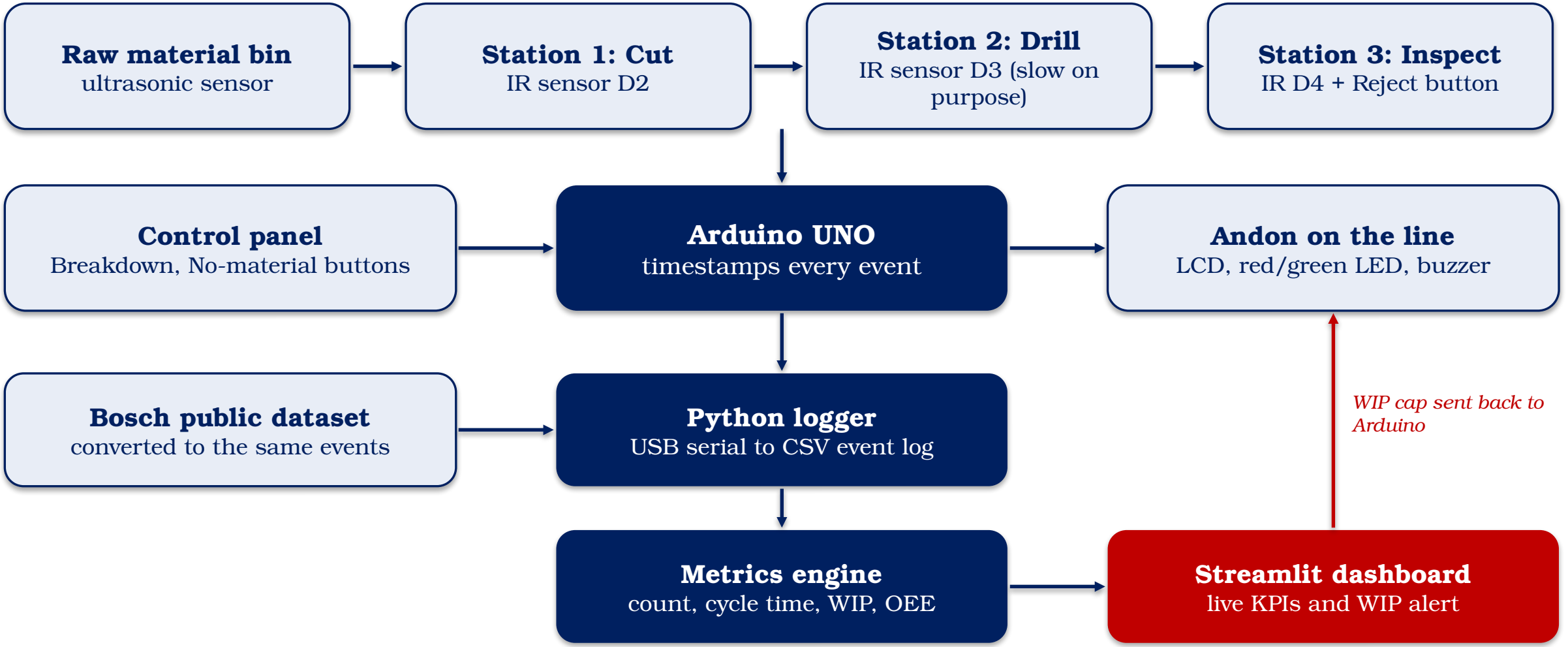
Methodology

- **Design:** quantitative and experimental. Build the line, measure it, change one thing at a time (rebalance, then WIP limit) and compare the numbers.
- **Data collection:** IR sensors time every part leaving each station; buttons log breakdown, no-material and reject; an ultrasonic sensor reads the raw material bin. The Arduino stamps each event in milliseconds and sends it over USB to a Python logger (CSV). Secondary data: the public Bosch production line dataset [8], [12].
- **Data analysis:** throughput, cycle time per station, WIP and flow time from the event log; Little's law as a cross-check [7]; OEE proxy = A × P × Q [3]; mean and spread across 3 repeat runs.
- **Tools:** Arduino IDE (C++), Python (pyserial, pandas, Streamlit), Excel for result tables.
- **Risks and fallbacks:** miscounts (cardboard gates, debounce, hand tally check); hardware failure on demo day (replay a saved run); Bosch time units are not published (use only for relative comparison).

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System Pipeline

Physical line (parts moved by hand)



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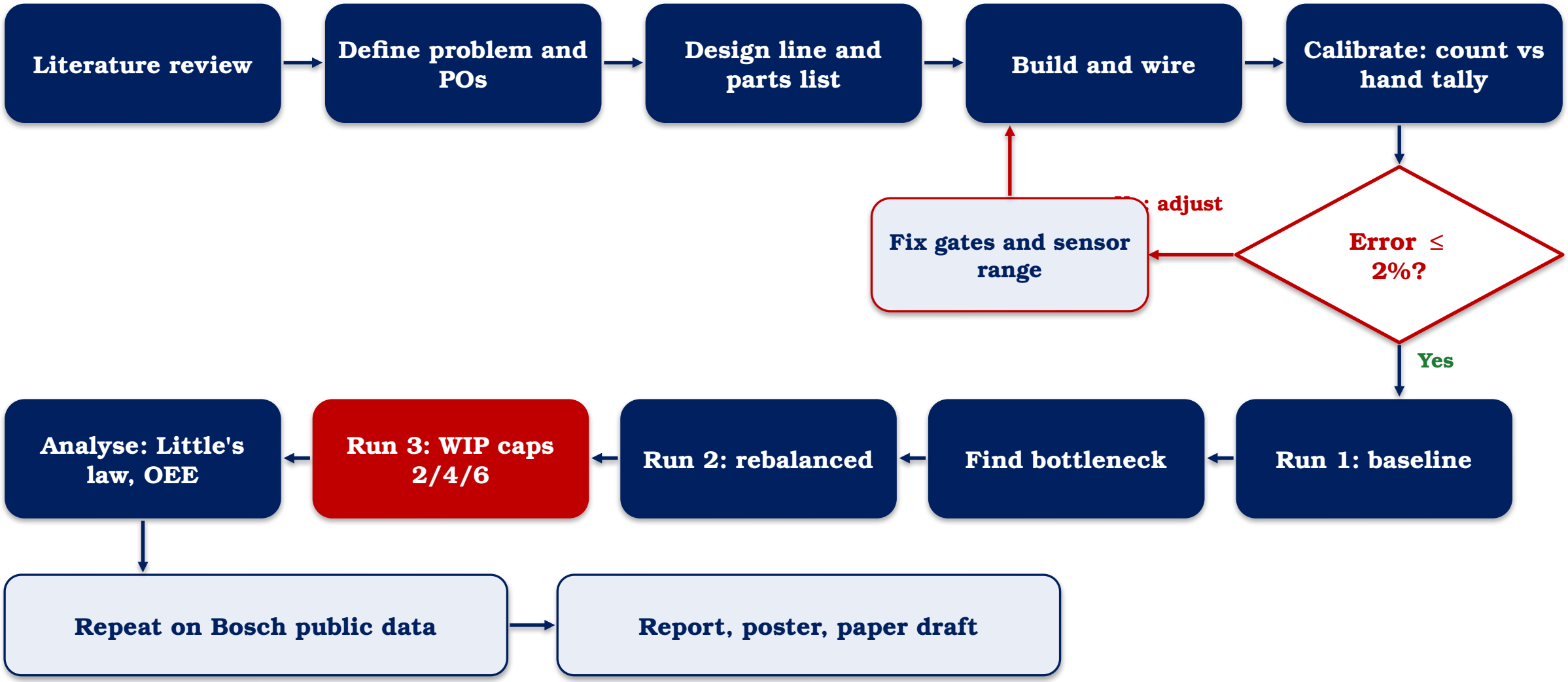
Measurement Parameters

No machine-learning model is trained in this project. These are the measurement settings we will report.

Item	Value
Data sources	Live line event log (CSV) and a Bosch sample: first 50,000 rows, line L3, 3 busiest stations [12]
Event format	time (s), event (PASS / SCRAP / DOWN / UP), station, reason
Sensor timing	IR debounce 400 ms; button debounce 200 ms; status sent every 1 s; dashboard refresh 2 s
Cycle time (per station)	finish(k) − max(arrive(k), finish(k−1)), so stations after the bottleneck do not look slow
WIP	parts released at Station 1 minus parts that left Station 3 (good or scrap)
Throughput / flow time	good parts per minute / exit time − entry time per part; checked with Little's law [7]
OEE proxy	A × P × Q [3]; ideal cycle time set by the team: __ **s**
Experiments	baseline, rebalanced, WIP cap 2 / 4 / 6; 10 minutes per run, 3 repeats each
Targets	count error ≤ 2%; Little's law gap ≤ 10%; alert fires every time WIP reaches the cap

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Research Process



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Proposed Outcomes

- **Expected results:** a working prototype (3-station line and live dashboard) and a results table comparing baseline, rebalanced and WIP-capped runs: throughput, average WIP, flow time, cycle time per station, OEE proxy and scrap rate.
- **Success criteria:** PO1 to PO4 met. Improvement after rebalancing: __ **%** (to be measured, not assumed).
- **Potential impact:** shows that count, cycle time, inventory and workflow can be watched live with low-cost parts, as a template for small units and student labs.
- **Limits we will state:** OEE is a proxy (our own ideal cycle time); parts are moved by hand; Bosch time units are estimated by the Kaggle community, not published by Bosch [8].
- **Future research directions:** wireless boards, more stations and product types, and testing the WIP-limit alert on more real factory data.

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Timeline of the Project

Sprint	Dates	Work	Linked to	Status
1	23 to 29 Oct 2026	Problem, literature review, POs, line design, parts list, code skeleton	Phase I review (29 Oct)	**Done**
2	30 Oct to 1 Nov 2026	Wire all sensors, logger and dashboard running together	PO1, PO2	**In progress**
3	2 Nov to __	Count accuracy test, Bosch data mode, first runs	PO1, PO2, Phase II	Not started
4	__ to __	Baseline, rebalance and WIP-cap experiments (3 repeats each)	PO3, PO4	Not started
5	__ to __	Results, report, poster, PPT, paper draft, demo rehearsal	Phase III	Not started

Phase II date: __ / __ / 2026 **Phase III date: __ / __ / 2027** (dates to be announced; sprints will be fixed backwards from them)

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Conclusion

- **Summary of key points:** OptiProductio measures production count, cycle time, inventory and workflow on a 3-station model line, live, using an Arduino UNO, low-cost sensors and a Python dashboard.
- **What makes it different:** it adds a WIP-limit alert based on CONWIP [5], and it checks its own numbers against a hand tally and Little's law [7].
- **Final thought:** you cannot improve a line you cannot see. OptiProductio makes the line visible cheaply.
- **Next steps:** finish wiring by 1 Nov, run the count accuracy test (PO1), then bring partial results to Phase II.

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Expected Outcome by the End of First Year

Product prototype: a working OptiProductio line and dashboard, released as open source (code, wiring table and build guide) so other labs can rebuild it.

Conference publication: a short paper in IEEE format on the prototype, the accuracy check and the WIP-limit experiments. Target venue to be chosen once results are in.

Research proposal: extend the work to more stations, wireless boards and more real factory data.

Patent: not planned. Software and algorithms on their own are generally not patentable in India (Patents Act, 1970, Section 3(k)) [13], so we will not claim one.


Thank You!